

A-1. Transforming Jhum into Argo-forestry



Agriculture Land Conservation



High



High



Terr

Measure Type

Cost

Preferences by Community

Hazards

Implementation Time of Measures

Descriptions

Measures	Jhum, or shifting cultivation is a traditional farming practice commonly used in hilly and forested areas which involves clearing forest land for agricultural use. Although a widely adopted agricultural practice in the hilly areas, it diminishes soil fertility and results in deforestation, soil erosion, and loss of biodiversity. Transforming Jhum into Agroforestry is a sustainable approach that integrates existing forest trees with agricultural crops on the same land, enhancing both agricultural productivity and environmental health.								
	Transformation from jhum to agroforestry can be accomplished through mixed cropping with tree planting where compatible agricultural crops are integrated with the existing forest trees. Alley cropping is another practice where trees are planted in rows with crops in the spaces between. These techniques support agricultural activities on hill slopes without posing threat to the existing forest tree species.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	Suitable for degraded lands, hill slopes, low forest								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
					√		√		
Feasibility criteria	- Choosing crop species compatible with the forest tree species and similar climate tolerance. - Selecting crop species having complementary soil nutrient requirement and water demand to the forest tree species. - Considering the slope and soil erosion risk of hilly terrain, contour planting or terracing can help mitigate erosion. - Considering the investment market return and profitability since forestry products may take years to materialize and result in profit. - Considering market access, value chain and demand of the forestry product as well as transportation and storage facilities.								
Operation and maintenance	Managing the soil nutrient and water demand between trees and crops as well as regular pruning and thinning is labor-intensive and requires proper knowledge of tree species and crop growth patterns.								
Cost	Medium								
Benefits									

	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			

Environmental performance	• Transforming jhum into argo-forestry integrates farming practices with forest landscape which increases agricultural production and contributes to conservation of landscape. • The integration of diverse species of trees, shrubs, and crops together form a complex and resilient ecosystem which promotes biodiversity, creating habitats for birds, insects, and other wildlife. • Trees contribute organic matter to the soil and enhance soil nutrient content. This helps maintain soil fertility over the long term, reducing the need for synthetic fertilizers. • The agroforestry system improves water retention in the soil through reducing surface water runoff and helping with groundwater recharge which reducing the risk of both drought and flooding.
Remarks	This measure will aid agricultural practices in CHT of Bangladesh without degrading the existing landscape.